

OBESITY

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Obesity

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Excess of adipose tissue deposition.

- Seen in >20% children in urban India.
- Energy intake >> expenditure → ↑↑ deposition of adipose primarily in the abdomen (Central Obesity : ↑↑ risk of complications).

Note :

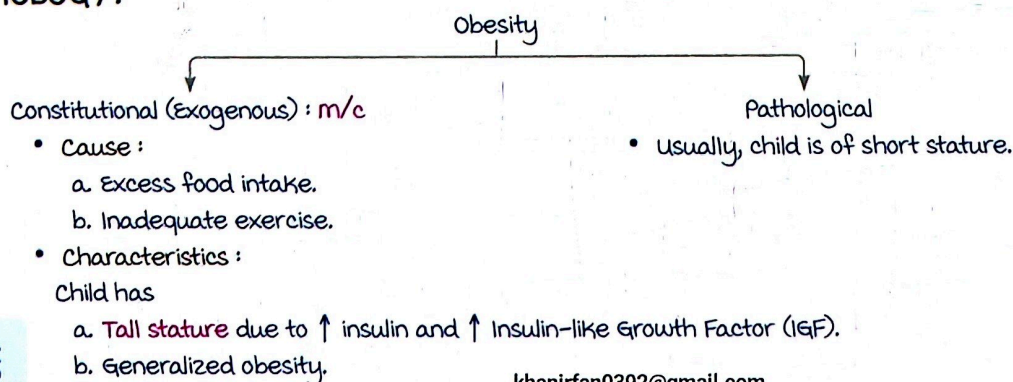
Body has unlimited capacity for storage of fats, but not carbohydrates or proteins.

DEFINITIONS :

Parameter	Measurement	Interpretation
• Body mass index : For children >5 yrs	• $BMI = \frac{\text{Weight (In kg)}}{\text{Height (In m}^2\text{)}} \times 100$ (Plotted in BMI charts)	Overweight : >23 adult equivalent Obesity : >27 adult equivalent
• Weight for height : For children <5 yrs	• WHO Weight for height charts.	• +2 to +3 SD : Overweight • >+3 SD : Obesity
• Skinfold thickness α ↑ subcutaneous fat	• Harpenden caliper.	>85th percentile : Obesity
• Waist circumference	• Indicator of abdominal obesity.	• > 75th percentile. • > 80 cm in girls. • > 90 cm in boys.
• Waist-hip ratio : Age independent	• Ratio of waist circumference to hip circumference.	>1 in males >0.85 in females
• Waist-height ratio : Age independent	-	>0.5 - obesity

Note : BMI interpretation is based on the Indian Association of Paediatrics BMI chart.

ETIOLOGY :



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Nutrition

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Causes of Pathological Obesity :**1. Monogenic causes :**

Hereditary single gene defects (Positive family history).

a. mineralocorticoid-4-receptor defect (mCAR) : m/c.

b. **Leptin deficiency** : ↓ leptin/resistance to leptin.

Uncontrolled appetite

Note : Leptin deficiency was the first reported monogenic cause of obesity.

2. Endocrine causes :

- Cushing's syndrome : Truncal obesity + buffalo hump + striae + moon-face.
- Hypothyroidism.
- Pseudohypoparathyroidism.
- Growth hormone deficiency.

3. Hypothalamic obesity :

- Lesion in **Ventro-medial nucleus (VMN)** : Satiety centre → Uncontrolled appetite.

• Causes :

- Head injury.
- Tumor.
- Radiation.
- Infection.

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• **ROHHAD** : mnemonic.

R : Rapid Onset.

H : Hypoventilation.

O : Obesity with.

A : Autonomic.

H : Hypothalamic dysfunction.

D : Dysregulation.

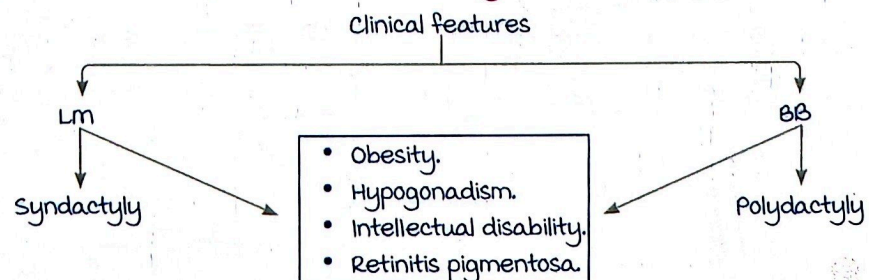
4. Syndromic obesity :a. **Prader-Willi Syndrome** : Genetic imprinting disorder.

- Clinical features : Hypogonadism, small hands & feet, almond-shaped eyes.

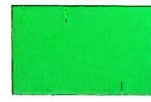
b. **Beckwith-Wiedemann syndrome** :

• Clinical features :

- Obesity
- Organomegaly.
- macroglossia
- Hemi hypertrophy.

c. **Lawrence-moon (Lm), Biedl-Bardet (BB) syndrome** :

Beckwith-Wiedemann syndrome



5. Drugs :

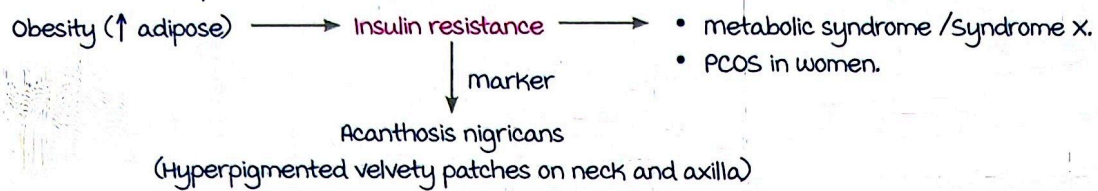
- Anti-epileptic drugs : E.g. Valproate, Lamotrigine.
- Steroids → Cushingoid appearance.

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Complications

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metabolic complications :



Non-Alcoholic Fatty Liver Disease (NAFLD) :

Leads to :

- Steatohepatitis (Inflammatory changes in liver) → Chronic → Fibrosis.

Management of obesity

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